Cambridge (CIE) A Level Physics

Your notes

Specific Heat Capacity & Specific Latent Heat

Contents

- Specific Heat Capacity
- Specific Latent Heat Capacity

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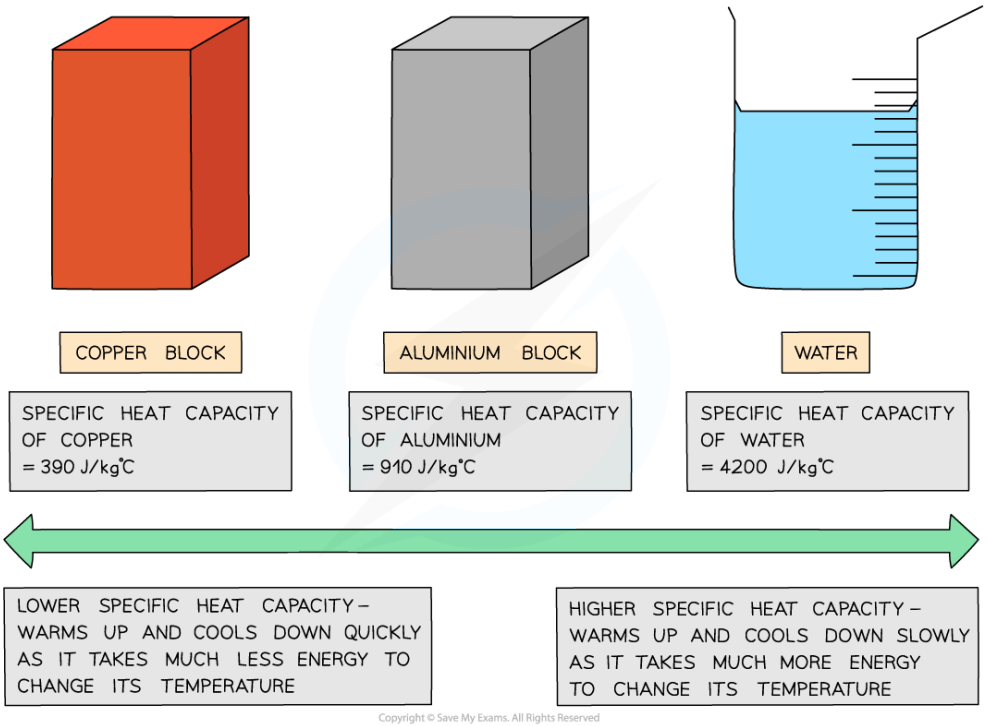
Defining specific heat capacity

- The specific heat capacity of a substance is defined as:

The amount of thermal energy required to raise the temperature of 1 kg of a substance by 1 °C

- This quantity determines the amount of energy needed to change the temperature of a substance
- The specific heat capacity is measured in units of **joules per kilogram per kelvin** ($J\,kg^{-1}K^{-1}$) or **joules per kilogram per Celsius** ($J\,kg^{-1}\,°C^{-1}$) and has the symbol **c**
  - Different substances have different specific heat capacities
  - Specific heat capacity is mainly used in liquids and solids

Specific heat capacity of different materials



Low v high specific heat capacity

- From the definition of specific heat capacity, it follows that:
  - The heavier the material, the more thermal energy that will be required to raise its temperature
  - The larger the change in temperature, the higher the thermal energy will be required to achieve this change

Calculating specific heat capacity



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- The amount of thermal energy Q needed to raise the temperature by $\Delta\theta$ for a mass m with specific heat capacity c is equal to:

$$\Delta Q = mc\Delta\theta$$

- Where:
  - ΔQ = change in thermal energy (J)
  - m = mass of the substance you are heating up (kg)
  - c = specific heat capacity of the substance ($J\,kg^{-1}\,K^{-1}$ or $J\,kg^{-1}\,°C^{-1}$)
  - $\Delta\theta$ = change in temperature (K or °C)
- If a substance has a **low** specific heat capacity, it heats up and cools down quickly (for equal masses heated at the same rate)
- If a substance has a **high** specific heat capacity, it heats up and cools down slowly (for equal masses heated at the same rate)
- The specific heat capacity of different substances determines how useful they would be for a specific purpose eg. choosing the best material for kitchen appliances

Table of values of specific heat capacity for various substances

Substance	Specific heat capacity / $J\,kg^{-1}\,K^{-1}$
Aluminium	910
Copper	390
Lead	126
Glass	500 - 680
Water	4200
Mercury	140

- Metals conduct thermal energy well because of their free electrons; this is a separate property from specific heat capacity



Worked Example

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A kettle is rated at 1.7 kW. A mass of 650 g of a liquid at 25°C is poured into a kettle. When the kettle is switched on, it takes 3.5 minutes to start boiling.

Calculate the specific heat capacity of the liquid.

Step 1: Calculate the energy from the power and time

$$\text{Energy} = \text{Power} \times \text{Time}$$

$$\text{Power} = 1.7\ \text{kW} = 1.7 \times 10^3\ \text{W}$$

$$\text{Time} = 3.5\ \text{minutes} = 3.5 \times 60 = 210\ \text{s}$$

$$\text{Energy} = 1.7 \times 10^3 \times 210 = 3.57 \times 10^5\ \text{J}$$

Step 2: Thermal energy equation

$$\Delta Q = mc\Delta\theta$$

Step 3: Rearrange for specific heat capacity

$$c = \frac{\Delta Q}{m\Delta\theta}$$

Step 4: Substitute in values

$$m = 650\ \text{g} = 650 \times 10^{-3}\ \text{kg}$$

$$\Delta\theta = 100 - 25 = 75\ ^\circ\text{C}$$

$$c = \frac{3.57 \times 10^5}{650 \times 10^{-3} \times 75} = 7323.07\ldots = 7300\ \text{J kg}^{-1}\,^\circ\text{C}^{-1}\ (2\ \text{s.f})$$

Examiner Tips and Tricks

The difference in temperature Δθ will be exactly the same whether the temperature is given in Celsius or kelvin. Therefore, there is no need to convert between the two since the **difference** in temperature will be the same for both units.

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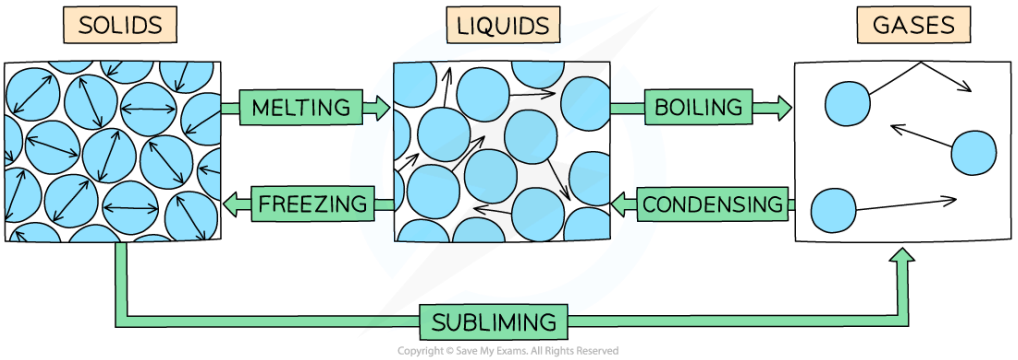
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Defining latent heat capacity

- Energy is required to change the **state** of a substance
- Examples of changes of state are:
  - Melting = solid to liquid
  - Evaporation/vaporisation/boiling = liquid to gas
  - Sublimation = solid to gas
  - Freezing = liquid to solid
  - Condensation = gas to liquid

Changes of state



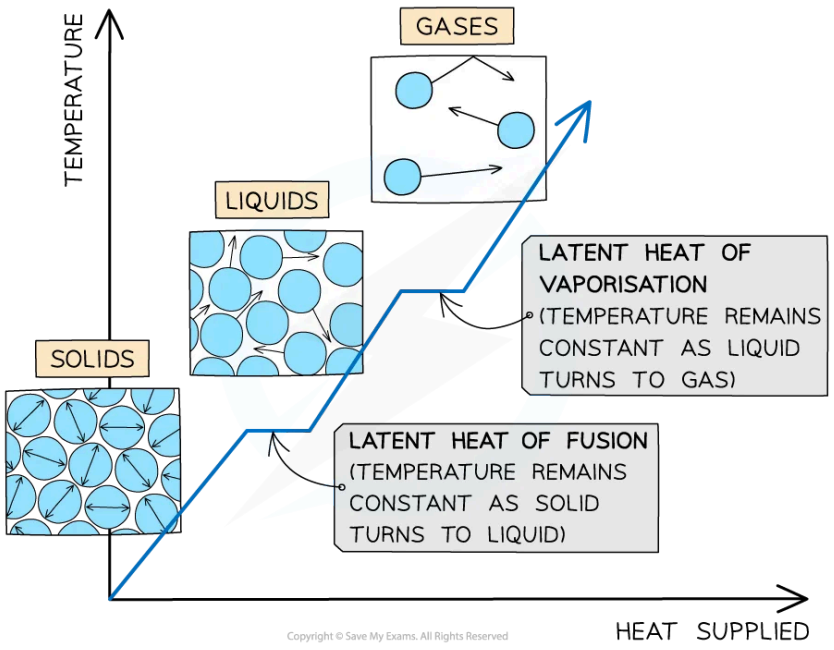
The example of changes of state between solids, liquids and gases

- When a substance changes state, there is **no temperature change**
- The energy supplied to change the state is called the **specific latent heat** and is defined as:

  The thermal energy required to change the state of 1 kg of mass of a substance without any change of temperature

- There are two types of latent heat:
  - Specific latent heat of **fusion** (melting)
  - Specific latent heat of **vaporisation** (boiling)
- It is important to recognise the difference between **latent heat** and **specific latent heat**
  - **latent heat** is the energy required to change a **materials state**
  - **specific latent heat** is the energy required to **change the state of 1 kg** of a material

Types of latent heat



The changes of state with heat supplied against temperature. There is no change in temperature during changes of state.

- The specific latent heat of **fusion** is defined as:

  The thermal energy required to convert 1 kg of solid to liquid with no change in temperature

- This is used when melting a solid or freezing a liquid
- The specific latent heat of **vaporisation** is defined as:

  The thermal energy required to convert 1 kg of liquid to gas with no change in temperature

- This is used when vaporising a liquid or condensing a gas

Calculating specific latent heat

- The amount of energy Q required to melt or vaporise a mass of *m* with latent heat *L* is:

$$Q = Lm$$

- Where:
  - Q = amount of thermal energy to change the state (J)
  - L = latent heat of fusion or vaporisation (J kg^{-1})
  - m = mass of the substance changing state (kg)
- The values of latent heat for water are:

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- Specific latent heat of fusion = 330 kJ kg^{-1}
- Specific latent heat of vaporisation = 2.26 MJ kg^{-1}

- Therefore, evaporating 1 kg of water requires roughly **seven times** more energy than melting the same amount of ice to form water
- The reason for this is to do with intermolecular forces:
  - **When ice melts**: energy is required to just increase the molecular separation until they can flow freely over each other
  - **When water boils**: energy is required to completely separate the molecules until there are no longer forces of attraction between the molecules, hence this requires much more energy

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Worked Example

The energy needed to boil a mass of 530 g of a liquid is 0.6 MJ.

Calculate the specific latent heat of the liquid and state whether it is the latent heat of vaporisation or fusion.

Step 1: Write the thermal energy required to change state equation

$$Q = Lm$$

Step 2: Rearrange for latent heat

$$L = \frac{Q}{m}$$

Step 3: Substitute in the values

$$m = 530\ g = 530 \times 10^{-3}\ kg$$

$$Q = 0.6\ MJ = 0.6 \times 10^{6}\ J$$

$$L = \frac{0.6 \times 10^{6}}{530 \times 10^{-3}} = 1.132 \times 10^{6}\ J\ kg^{-1} = 1.1\ MJ\ kg^{-1}\ (2\ s.f.)$$

L is the latent heat of vaporisation because the change in state is from liquid to gas (boiling)

Examiner Tips and Tricks

Use these reminders to help you remember which type of latent heat is being referred to:

- Latent heat of fusion = imagine 'fusing' the liquid molecules together to become a solid

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- Latent heat of vaporisation = “water vapour” is steam, so imagine vaporising the liquid molecules into a gas



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